SOCIETY ENGINE / OPEN CONTRIBUTION / GUARDED PRODUCTION

Society Engine Protocol

Risk-managed architecture for a self-growing Web3 game economy.



ZTAKORA

PROTOCOL ARCHIVE / V1.0

EDITION
1.0

STATUS
DESIGN BASELINE

ISSUED
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COMPANY CONFIDENTIAL

CONTROL

Document Control

Authority, use and interpretation of this protocol baseline.

Field	Value
Document title	Ztakora Society Engine Protocol v1.0
Purpose	Define a realistic, risk-managed protocol architecture for an open and continuously expanding Web3 game society.
Primary audience	Ztakora leadership, product, game, blockchain, security, legal and operating teams; selected development and investment partners.
Decision use	Architecture alignment, vertical-slice planning, staffing estimation, legal scoping and Phase 1 approval.
Classification	Company Confidential - Internal Design Baseline
Supersedes	Earlier mining-led and multi-DAO concepts

INTERPRETATION

This document is a product and protocol specification, not a promise of a risk-free system. Its architecture removes or contains the most dangerous failure paths: speculative token issuance, fake DAO activity, automatic production deployment, unrestricted dependencies and unverified investment products.

DOCUMENT NAVIGATION / SOCIETY ENGINE PROTOCOL

26 ENTRIES / 06 SECTION GROUPS

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A structured index to the protocol's decisions, controls, implementation path and reference frameworks.

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OVERVIEW

Executive Summary

Ztakora Society Engine Protocol v1.0 (ZSEP) is a decentralized creation and company-formation layer for a shared 3D game world. It allows any participant to identify a societal need, publish a product concept, assemble a builder team and test a working service. The protocol then admits only demonstrably useful, secure and maintainable products into the canonical world.

The central design decision is to decentralize opportunity without decentralizing production safety. Idea creation and prototype formation remain open. Main-world deployment, token activation, cross-product integration and economic permissions remain conditional on published technical, creative, economic and operational gates.

> PROTOCOL THESIS
> Growth should be earned through useful player adoption, not through mining, token emissions, raw votes, transaction count or speculative capital.

Executive decisions embedded in v1.0

- The protocol runs on an established EVM network; it does not introduce a new validator or miner consensus.
- There is one constitutional Protocol DAO. Domains are registries, and companies use bounded Company Contracts rather than mandatory DAOs.
- New products launch first in an isolated Innovation District and cannot alter the canonical economy during testing.
- Proof of Useful Adoption (PoUA) replaces vote-only approval. A working prototype and genuine, unrewarded player use are required before graduation.
- The ecosystem uses one settlement asset. Company-specific tradable ERC-20 tokens are not a default entitlement.
- Service tokens and functional NFTs are created from audited factories and must encode actual game utility, supply limits and recovery controls.
- Permissionless prototype backing is separated from regulated investment involving equity, revenue rights or expected financial return.
- Every live product carries a maintenance reserve, continuation license, dependency declaration and controlled retirement path.

Protocol at a Glance

How an open need becomes a bounded canonical service.

PROTOCOL LOGIC / GUARDED PROGRESSION

01 **OPEN NEED**

Anyone may identify a specific unmet player need.

OPEN ENTRY

02 **SANDBOX BUILD**

Teams prototype in an isolated environment with bounded permissions.

SAFE FAILURE

03 **POUA GATE**

Independent evidence tests function, adoption, integrity, safety and maintainability.

EVIDENCE

04 **CANONICAL SERVICE**

Useful products earn scoped production privileges and remain recoverable.

CONTROLLED ACCESS

Opportunity is open. Production safety remains conditional on published evidence and bounded permissions.

Foundations

Sections 01-04 / Product thesis, governing principles, participants and protocol architecture



01 Strategic Context and Product Thesis

1.1 The problem with studio-only virtual worlds

Traditional online games are centrally produced. A studio defines the economy, regions, professions, assets and content cadence. This protects quality, but it limits the rate and diversity of expansion. Web3 games often add ownership but retain the same studio-controlled production model; alternatively, they open token issuance and NFT creation too quickly, producing speculation without durable gameplay demand.

ZSEP addresses the missing middle. It creates an open production market while preserving a canonical layer that players can trust. The protocol does not assume that decentralization automatically produces quality. It creates bounded places where experimentation can fail safely and evidence-based gates through which useful products can earn greater privileges.

1.2 Society as a product-generation model

Real economies grow because participants encounter needs, organize resources, form ventures, build products and exchange value. ZSEP reproduces that coordination pattern inside a game world without pretending that an on-chain organization is equivalent to a legal company. Game-native companies can operate transportation, housing, crafting, entertainment, mapping, logistics, communication and other services. Their products must remain subordinate to the game constitution and to player enjoyment.

1.3 Product thesis

> THESIS
> A self-growing game ecosystem becomes viable when the protocol makes creation open, experimentation cheap, failure contained, adoption measurable, ownership legible and maintenance economically mandatory.

1.4 What remains on-chain and off-chain

REFERENCE TABLE

On-chain control plane	Off-chain execution plane
Proposal identity and status	High-frequency gameplay simulation
Milestone escrow and treasury permissions	3D rendering, physics and animation
Company charters and role assignments	Large asset files and build artifacts
Token supply rules and asset provenance	Telemetry processing and anomaly analysis

On-chain control plane	Off-chain execution plane
Revenue routing and maintenance reserves	Moderation evidence and creative review
Governance decisions and timelocks	Search, discovery and player recommendations

This division keeps blockchain functions auditable while avoiding the cost and latency of placing real-time game state or large 3D content on-chain.

02 Design Principles and Non-Goals

2.1 Governing principles

REFERENCE TABLE

Principle	Required interpretation
Open entry	Any participant may publish a draft idea, join a team or create a prototype without purchasing governance power.
Evidence over popularity	A product advances because it works and players return to it, not because a token holder majority likes a description.
Safe failure	Experimental products are isolated from canonical assets, balances and critical dependencies until graduation.
Minimum necessary governance	Governance defines standards and exceptions; it does not vote on routine operations or every product decision.
One stable economic base	The protocol begins with one settlement asset and tightly controlled resource representations.
Utility before liquidity	Tokens are activated only when the associated product is usable and the token performs a necessary function.
No pay-to-win	Purchasable assets may provide expression, access or bounded utility but cannot sell unrestricted combat dominance.
Separation of duties	Proposal, implementation, verification, treasury control and emergency intervention cannot be concentrated in one actor.
Maintenance by design	Revenue reserves, continuation rights and retirement procedures exist before production launch.
Progressive decentralization	Ztakora can operate defined safety and creative functions initially, with transparent migration criteria rather than false day-one autonomy.

2.2 Explicit non-goals

- Creating a new layer-1 blockchain, mining network or validator marketplace.
- Rewarding participants for raw votes, transactions, wallet creation, asset count or validator volume.
- Creating one freely tradable token for every company, region or market.
- Allowing AI, telemetry or a single DAO vote to deploy arbitrary contracts directly into production.
- Offering guaranteed yield, redemption, appreciation, dividends or venture returns through the permissionless protocol.
- Placing all gameplay, 3D files or player behavior data on-chain.
- Removing professional creative direction, security review, moderation or legal accountability.
- Supporting cross-chain bridges during the initial operating stage.

03 Ecosystem Participants and Value Exchange

ZSEP is a multi-sided production network. Its integrity depends on participants having distinct responsibilities and on rewards being tied to delivered, maintained utility rather than promotional activity.

REFERENCE TABLE

Participant	Primary responsibility	Permitted reward
Player / citizen	Identifies needs, tests products, uses services and contributes bounded governance signals.	Gameplay value, access, reputation and optional community privileges.
Proposer	Defines a problem, target users, concept, acceptance measures and failure conditions.	Attribution and a capped usage royalty only if a shipped product creates real value.
Economy designer	Models sources, sinks, prices, dependencies, scarcity and abuse cases.	Milestone payment and approved maintenance work.
Smart contract developer	Implements Company Contracts, token modules, escrow and integration adapters.	Milestone payment and bounded maintenance or usage compensation.
Game developer	Builds client, server and gameplay integration.	Milestone payment and maintenance revenue.
3D creator	Produces licensed, optimized and versioned functional assets.	Production payment and bounded asset royalty.
Backer	Funds a prototype or provides non-financial resources.	Access, service credits, cosmetics or regulated rights outside the permissionless layer.
Verifier	Reviews code, assets, economics and evidence against public requirements.	Fixed fee independent of pass or fail; reputation based on review quality.
Operator	Runs support, monitoring, maintenance and customer operations.	Service revenue subject to reserve and performance obligations.
Protocol steward	Maintains standards, emergency controls and constitutional processes.	Transparent operating budget; no unilateral ownership seizure.

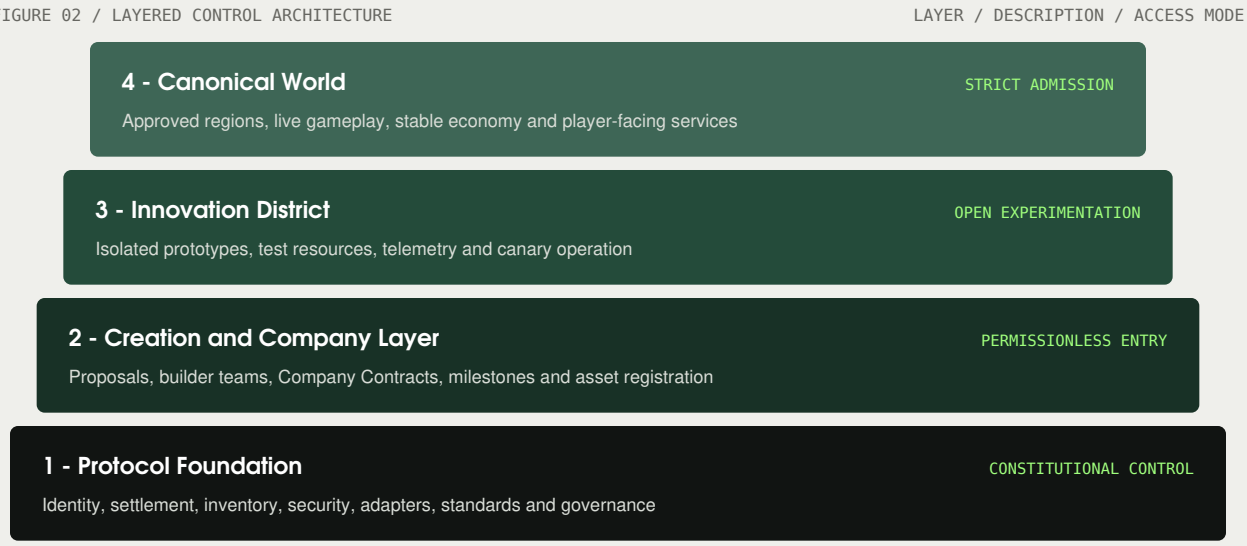
INCENTIVE RULE

Reviewers are paid for completing valid reviews, not for approving projects. Players are not paid for raw votes. Proposers earn only after useful adoption. These rules reduce approval bias, vote farming and proposal spam.

04 Protocol Architecture

Architecture at a glance. Each layer has a deliberately limited responsibility.

FIGURE 02 / LAYERED CONTROL ARCHITECTURE



4.1 Protocol Foundation

The foundation is the smallest trusted layer. It provides identity and reputation interfaces, one settlement rail, canonical inventory references, Company Contract templates, proposal status, token and NFT factories, revenue routing, governance timelocks, emergency isolation and dependency adapters. Foundation contracts should change rarely and only through audited, timelocked releases.

4.2 Creation and Company Layer

This layer converts ideas into bounded organizations. It contains the Proposal Registry, team formation, milestone escrows, Company Contracts, role credentials and product manifests. It is open to new participants but cannot bypass the Innovation District or canonical admission gates.

4.3 Innovation District

The Innovation District is an isolated test environment connected to the game client. It uses test balances, regional permissions, rate limits and explicit risk labels. Product contracts cannot debit canonical balances or invoke critical services without adapters. Players opt in and understand that products may reset, fail or be removed.

4.4 Canonical World

The Canonical World contains approved regions, stable gameplay systems and products that have passed technical, creative, economic, legal and operational gates. Graduation grants scoped capabilities rather than unrestricted access. A transportation company, for example, may receive permission to charge approved fares and move defined cargo types, but not to mint land or modify combat attributes.

4.5 Control boundary

REFERENCE TABLE

Action	Open by default	Conditional	Prohibited
Publish an idea	Yes	Formal qualification requires complete fields and an anti-spam bond.	Impersonation, copied content or concealed conflicts.
Build a prototype	Yes	Must use sandbox permissions for integrated testing.	Direct canonical contract access.
Create functional NFT	Yes, in sandbox	Canonical registration requires manifest and validation.	Silent content replacement.
Create service token	No	Factory activation after utility and economic review.	Unlimited administrator minting.
Enter Canonical World	No	PoUA plus safety, creative and operating gates.	Vote-only or payment-only admission.
Offer financial return	No	Separate regulated module and legal entity.	Permissionless guaranteed return.

II

Process & Governance

Sections 05-08 / Proposal lifecycle, useful adoption, company operation and decision rights



05 Proposal-to-Product Lifecycle

The lifecycle is a state machine. Each transition has an owner, evidence requirement, financial effect and reversal path. Smart contracts enforce objective transitions; human reviewers sign subjective or professional determinations.

LIFECYCLE STATE MACHINE / REQUIRED OUTPUTS AND CONTROLS

State	Primary activity	Required output	Control
0. Need signal	Player, company or analyst documents a specific unmet need.	Problem statement and affected user group.	No funds or tokens.
1. Draft proposal	Anyone publishes a structured concept.	Scope, users, dependencies, budget range and success criteria.	Editable and withdrawable.
2. Qualification	Completeness, originality and basic feasibility are checked.	Formal proposal record and refundable bond.	Reject spam; no product approval implied.
3. Team formation	Builders voluntarily commit roles and delivery capacity.	Named responsibilities, conflicts and preliminary architecture.	No team assignment by vote.
4. Prototype funding	A capped milestone vault is funded.	First build budget and public milestone tests.	Funds released only for verified work.
5. Sandbox build	Team implements inside the Innovation District.	Working prototype, source package, asset manifest and test results.	No canonical balances or dependencies.
6. Adoption trial	Unrewarded players use the product for a defined period.	Cohort, retention, reliability and satisfaction evidence.	Anomalous activity excluded or challenged.
7. Admission review	Independent specialists evaluate required gates.	Signed technical, creative, economic and operational findings.	Conflicted reviewers are disqualified.
8. Canary launch	Product enters one bounded canonical region.	Limited permissions, budgets, users and dependencies.	Automatic circuit breakers and rollback.
9. Graduation	Sustained performance earns normal scoped operation.	Production Company Contract and maintenance reserve.	Privileges remain revocable for defined breaches.
10. Evolution	Company proposes upgrades, new regions or products.	Versioned extension proposal and compatibility evidence.	No silent charter expansion.
11. Recovery / retirement	Failed or abandoned product is transferred or wound down.	Customer protection, asset continuity and final accounting.	No forced asset seizure.

5.1 Proposal specification

- Problem and affected player segment
- Proposed product, service or infrastructure
- Expected gameplay value and player journey
- Required smart contracts, game services and 3D assets
- Economic inputs, outputs, sources, sinks and pricing assumptions
- Dependencies and required fallback behavior
- Prototype budget, milestones and public acceptance tests
- Content ownership, licenses and AI-generation disclosure
- Operational owner, support plan and maintenance estimate
- Success, suspension, takeover and shutdown conditions

5.2 Formal qualification does not equal approval

Qualification means that a proposal is complete enough to evaluate and discover. It does not grant treasury funding, a company token, canonical deployment or a claim of endorsement. This distinction is essential: the protocol should make it inexpensive to express ideas but expensive to misrepresent readiness.

06 Proof of Useful Adoption

Proof of Useful Adoption (PoUA) is an application-level admission mechanism, not block consensus. The underlying chain retains its native proof-of-stake security. PoUA determines whether a game product has earned expanded operating permissions.

6.1 Required proof categories

REFERENCE TABLE

Proof	Question answered	Representative evidence
Need	A documented problem exists for a defined player cohort.	Independent need reports, search or support patterns, existing service failure.
Function	A working prototype completes its core user journey.	Recorded tests, reproducible build, sandbox contract state and client integration.
Adoption	Players use the product voluntarily and return.	Unique cohorts, retention, task completion, repeat use and satisfaction.
Integrity	Usage is not predominantly controlled, rewarded or purchased by the team.	Cohort diversity, relationship analysis, device/rate limits and challenge review.

Proof	Question answered	Representative evidence
Safety	Code, assets and operations stay within published limits.	Security tests, performance budgets, moderation and incident readiness.
Economic compatibility	The product does not destabilize scarcity, progression or adjacent services.	Simulation, capped exposure, price/resource sensitivity and canary evidence.
Maintainability	A responsible operator and sustainable reserve exist.	Support owner, maintenance budget, continuation license and recovery plan.

6.2 Adoption metrics

No single metric determines graduation. Thresholds depend on product risk, cost and expected frequency. A high- frequency marketplace requires different evidence than a monthly festival venue or a regional bridge. The Protocol DAO approves metric classes and ranges; project teams publish the exact threshold before the trial begins.

REFERENCE TABLE

Metric	Interpretation	Gate form
Independent users	Number and diversity of eligible testers after excluding team-controlled and rewarded accounts.	Minimum cohort by risk class.
Return rate	Share of eligible users returning within the product's natural use cycle.	Cohort-adjusted threshold.
Task completion	Share completing the intended utility without support intervention.	Published journey target.

REFERENCE TABLE

Metric	Interpretation	Gate form
Unrewarded usage	Use not generated by token, vote or test incentives.	Required majority of qualifying activity.
Reliability	Successful service interactions and error-free sessions.	Higher threshold for critical dependencies.
Player value	Satisfaction, time saved or new gameplay enabled.	Survey plus behavioral evidence.
Concentration	Share of activity, supply or revenue controlled by top related parties.	Constitutional maximums.
Economic impact	Change to prices, scarcity, progression and adjacent companies.	Within predeclared guardrails.

6.3 Anti-manipulation controls

- Testing rewards are labeled and excluded from adoption qualification.

- Related wallets, shared funding sources and coordinated behavior are investigated as evidence, not treated as automatic identity truth.
- Metric definitions, trial dates and exclusion rules are committed before data collection.
- Randomized player cohorts supplement self-selected testers for higher-risk products.
- Reviewers can challenge a cohort; the team receives a documented response and appeal path.
- Anomaly models cannot independently reject or approve a product; they trigger human review.

07 Company Contract and Operating Model

A Company Contract is a bounded on-chain operating charter. It provides transparent roles, permissions and financial controls without requiring every product team to operate a full political DAO. The company may be a game-native organization, a conventional legal entity or a connected pair of both.

7.1 Required modules

REFERENCE TABLE

Module	Responsibility
Charter	Purpose, product scope, permitted regions, capabilities and prohibited actions.
Role registry	Proposer, builders, operator, treasury signers, reviewers and emergency contacts.
Milestone vault	Fund custody, objective release conditions, disputes, refunds and closeout.
Permission controller	Scoped calls into settlement, inventory, spatial and event adapters.
Product registry	Contract versions, asset manifests, dependencies, service endpoints and status.
Revenue router	Disclosed allocations for operation, creators, maintenance and protocol fees.
Maintenance reserve	Restricted funds for patches, support, compatibility and customer continuity.
Upgrade controller	Timelocks, canary releases, compatibility tests and rollback references.
Recovery controller	Warning, isolation, takeover, migration, retirement and final accounting.

7.2 Permission model

Permissions should be capability based. A company receives only the minimum functions required for its product. Permissions carry ceilings for value, frequency, geography, inventory category and contract destinations. A generic administrator role with unrestricted external calls is not acceptable.

7.3 Revenue policy

Each company publishes its revenue route before launch. The protocol sets constitutional boundaries rather than a universal split. A representative configuration may allocate 65-75 percent to operations and builders, 10-20 percent to mandatory maintenance, 5-15 percent to approved asset creators and no more than 5 percent to the protocol. The exact route must be visible at the point of purchase.

7.4 Company evolution

A company may add regions, products or economic rights only through a versioned charter extension. Routine bug fixes and compatible asset updates follow an operating release process; material changes to supply, pricing power, player rights or dependencies require renewed review and canary operation.

08 Governance and Decision Rights

ZSEP uses one constitutional Protocol DAO. It governs the rules of the market, not the daily decisions of every participant. Domain labels organize discovery but do not create separate treasuries or voting systems by default.

REFERENCE TABLE

Decision	Primary authority	Constraint
Protocol constitution	Protocol DAO	Supermajority, timelock and public challenge.
Routine company operation	Company operator and bounded multisig	Within charter and automated limits.
Prototype design	Builder team	Open specification and sandbox restrictions.
Technical admission finding	Independent verifier set	Published checklist and conflict disclosure.
Creative admission finding	Rotating game/creative reviewers	Regional style and lore standard.
Economic admission finding	Economy reviewers	Simulation, concentration and balance guardrails.
Emergency isolation	Limited guardian multisig	Pause only; no asset seizure; public reason; automatic expiry.
Dispute appeal	Randomized appeal panel	New reviewers, evidence record and bounded timeline.

8.1 Governance power

- Governance participation uses unique-person credentials and earned, non-transferable reputation rather than purchased token weight.
- Delegation expires and must be renewed, preventing permanent invisible control.

- Reputation decays when participants stop contributing and cannot guarantee permanent office.
- Sensitive votes use commit-reveal where front-running or coordination pressure is material.
- Conflicts, related entities and paid representation must be disclosed for proposal-specific decisions.
- Constitutional changes cannot move user assets, retroactively alter ownership or waive legal obligations.

8.2 Progressive decentralization

During the initial product stage, Ztakora should retain responsibility for the base game, canonical creative direction, security release process and legal perimeter. These responsibilities must be explicit, narrow and measurable. Transfer to broader governance occurs only after operational maturity, verifier diversity, tested incident response and sufficient independent participation.

III

Economics & Assets

Sections 09-12 / Settlement, economic stability, funding separation and 3D content standards



09 Token, NFT and Settlement Architecture

The token model is intentionally smaller than the company model. A company can exist and charge for a service without issuing its own freely traded token. This removes a major source of speculation, fragmented liquidity and regulatory ambiguity.

REFERENCE TABLE

Instrument	Purpose	Core restriction
Settlement asset	Payments across approved products.	One stable-value asset or closed game credit; no company control over supply.
Participation reputation	Governance eligibility and contribution history.	Non-transferable, scoped, challengeable and decaying.
Service credit	Prepaid access or unit of a defined service.	Capped issuance, disclosed redemption rules, no governance rights.
Resource token	Consumable game input or output.	Minting tied to approved production; sinks and supply ceilings required.
Functional NFT	Ownership or licensed use of a specific item, structure or vehicle.	Versioned manifest, bounded attributes and canonical compatibility status.
Company charter identity	Canonical identity and status of a company.	Non-transferable; not equity; records permissions and versions.
Funding receipt	Evidence of a contribution to a milestone vault.	Does not automatically convey profit, equity or tradability.

9.1 Token factory requirements

- Fixed maximum supply or a constitutionally bounded supply function.
- Role-based minting with timelock, per-period ceiling and public event reason.
- Defined consumption, burn, expiry or service redemption mechanism where appropriate.
- No unlimited owner mint, concealed blacklist or arbitrary confiscation function.
- Emergency pause limited to the affected token or product.
- Transparent initial allocation, vesting and related-party disclosure.
- No transferability until the associated product is operational and transfer serves a documented use.
- Simulation and canary limits before canonical economic exposure.

9.2 NFT meaning

An NFT records identity, provenance, ownership or license state. It does not place a complete 3D model on-chain, guarantee permanent game support or convey intellectual-property rights that the creator does not possess. The user interface must disclose the exact rights, dependencies, version and canonical status before purchase.

10 Economic Stability and Game Balance

The world economy is a shared system. A locally successful company can still damage the wider game by eliminating scarcity, bypassing progression or concentrating essential infrastructure. Every product therefore receives an economic exposure class and explicit guardrails.

10.1 Required economic model

REFERENCE TABLE

Model area	Required question
Demand	Who needs the product, why, how often and what alternative exists?
Inputs	Which resources, labor, assets, locations or permissions are consumed?
Outputs	Which goods, services, access rights or advantages are produced?
Sources and sinks	How does value enter and permanently leave the system?
Supply control	What caps production, inventory, throughput and related-party concentration?
Pricing	What is fixed, market-priced, auctioned or adjusted, and who may change it?
Substitution	Can another supplier or fallback recipe serve the same need?
Failure impact	What happens to customers and dependencies if the service pauses?
Progression impact	Does the product bypass skill, time, exploration or cooperative play?

10.2 Guardrails

- No product may sell unrestricted raw combat power or permanent immunity.
- Essential services require at least one fallback path and may not receive permanent exclusive rights.
- Production and minting are capped by region, period and related-party concentration.
- Scarcity-sensitive changes use canary exposure and rollback thresholds.
- Revenue cannot be sustained by permanent protocol subsidies after the defined incubation period.
- Players must see fees, expiration, maintenance dependency and resale restrictions before purchase.

- Starter regions and catch-up mechanics prevent early companies from permanently excluding new players.

10.3 Competition and interoperability

Canonical interfaces should support multiple suppliers. Shared recipes, routes and critical services require fallback providers or protocol-operated emergency substitutes. Company-specific advantages may exist, but no product should obtain an irreversible monopoly over a fundamental player action.

11 Funding, Investment and Legal Separation

ZSEP distinguishes product backing from financial investment. This is a structural separation, not a change of terminology. A contribution that creates an expectation of profit may remain regulated even if it is called a patron receipt, NFT or DAO token.

11.1 Permissionless prototype backing

The base protocol may accept milestone funding for development where the participant receives access, service credits, cosmetic assets, public recognition or participation privileges. Terms must state that delivery can fail, credits may be game-limited and no financial return is guaranteed.

11.2 Regulated commercial investment

Equity, revenue participation, dividends, debt, profit-sharing or transferable instruments with an investment expectation belong in a separate jurisdiction-specific module connected to a legally accountable entity. The game protocol can reference the entity and its Company Contract but should not automatically distribute investment instruments to anonymous wallets.

REFERENCE TABLE

Area	Permissionless backing	Regulated investment lane
Access	Open where no financial return is offered.	Eligibility, KYC and regional controls as legally required.
Instrument	Service credit, cosmetic, access right or acknowledgment.	Equity, debt, revenue right or other regulated claim.
Custody	Milestone escrow with refund and dispute rules.	Approved custody, banking and legal- accounting process.
Disclosure	Product scope, delivery risk and credit limitations.	Financial risk, issuer, investor rights, tax and jurisdiction disclosures.
Marketing	Product utility and participation.	No public promise of yield, appreciation or guaranteed liquidity.

LEGAL BOUNDARY

DAO governance, decentralization, a utility label or an NFT wrapper does not remove securities, consumer, tax, sanctions, money-transmission or data-protection obligations.

12 3D Asset and Content Standard

A functional NFT must resolve to a versioned Game Asset Manifest. The manifest makes the asset usable, reviewable and recoverable across game-client releases. The chain stores the manifest hash and rights state; large models, textures, audio and animation packages remain in approved content storage.

REFERENCE TABLE

Manifest group	Required information
Identity	Asset ID, collection, company, creator, version and canonical status.
Content	Content-addressed package hashes, model formats, textures, audio and animation references.
Runtime	Engine version, client compatibility, collision, physics, dimensions and spatial anchors.
Performance	Polygon, draw-call, texture, memory, storage, CPU, bandwidth and LOD budgets.
Gameplay	Attributes, permitted modifiers, interactions, recipes, regions and dependency interfaces.
Rights	Ownership, commercial license, resale limits, royalty route and jurisdiction restrictions.
Provenance	Source files, creator attestations, third-party components and AI-generation disclosure.
Safety	Moderation status, malware/script scan, security findings and quarantine state.
Lifecycle	Upgrade history, supported versions, maintenance owner, fallback rendering and retirement path.

12.1 Content zones

Innovation Districts may support broad styles and experimental mechanics. Canonical regions publish visual profiles covering architecture, materials, scale, lighting, technology level, lore, cultural context and performance budgets. A technically valid asset can still be rejected from a specific canonical region while remaining available in an experimental or compatible region.

12.2 Versioning and buyer protection

- Material asset changes create a new manifest version and visible change log.
- Owners receive advance notice when an update changes appearance, functionality or compatibility.
- A creator cannot silently replace approved content with unrelated, malicious or lower-quality files.

- Fallback representations preserve inventory legibility if a detailed asset becomes unavailable.
- Royalty, license and dependency changes cannot be applied retroactively without an approved migration path.

IV

Technical & Risk

Sections 13-15 / Smart contracts, security / privacy, operations and recovery



13 Technical and Smart Contract Architecture

13.1 Recommended deployment posture

ZSEP should begin as an application protocol on a mature EVM layer-2 environment. Chain selection should consider security inheritance, stablecoin availability, transaction cost, account abstraction support, ecosystem tooling, operational monitoring and bridge dependence. The first release should use one chain and avoid protocol- owned bridges.

REFERENCE TABLE

Component	Primary responsibility
ProposalRegistry	Creates immutable proposal IDs, versions, status and evidence references.
QualificationBond	Holds refundable anti-spam bonds and applies narrow forfeiture rules.
MilestoneVault	Custodies funding, release approvals, disputes, refunds and closeout.
CompanyFactory	Deploys bounded Company Contracts from approved templates.
RoleRegistry	Stores scoped company and verifier credentials with expiry and revocation.
CapabilityController	Limits contract calls by function, destination, value, frequency and region.
AssetRegistry	Records manifest hashes, licenses, versions, status and canonical compatibility.
TokenFactory	Deploys approved service/resource tokens with enforced supply controls.
RevenueRouter	Splits payments and funds restricted maintenance reserves.
DependencyRegistry	Declares interfaces, providers, versions, fallbacks and health status.
GovernanceTimelock	Queues constitutional and high-impact changes for public review.
EmergencyIsolator	Pauses an affected capability or product without seizing unrelated assets.

13.2 Upgrade policy

Core registries and ownership records should minimize upgradeability. Where upgradeable components are necessary, upgrades require a versioned implementation allowlist, public diff, independent review, timelock, canary deployment and rollback reference. Company contracts cannot upgrade foundation contracts or bypass the capability controller.

13.3 Game integration adapters

- Settlement adapter for approved payments and refunds

- Inventory adapter for canonical item references and ownership changes
- Spatial adapter for placement, region and capacity permissions
- Event adapter for quests, sessions, transport, commerce and world events
- Identity adapter for privacy-preserving eligibility attestations
- Telemetry attestation adapter for signed aggregate adoption evidence

Adapters provide a stable boundary between slower on-chain control and faster game execution. They must enforce replay protection, rate limits, version compatibility and explicit failure behavior.

14 Security, Integrity and Privacy

Security is treated as a set of enforced boundaries rather than an audit performed once before launch. Every product receives a risk class based on value at risk, privilege, dependency depth, asset mutability and player harm potential.

REFERENCE TABLE

Threat	Minimum control set
Smart contract exploit	Audited templates, invariant and fuzz tests, bounded permissions, timelocks, canary limits and targeted pause.
Treasury theft	Milestone vaults, multisignature controls, withdrawal ceilings, allowlists and delayed high-value settlement.
Sybil / fake demand	Unique credentials, unrewarded cohorts, related-party analysis, rate limits and reviewer challenge.
Governance capture	Non-transferable reputation, expiring delegation, quorum diversity, conflict disclosure and timelocks.
Malicious asset	Content-addressed packages, scanning, script sandboxing, moderation, manifest versioning and quarantine.
Dependency failure	Adapters, circuit breakers, health monitoring, fallback providers and graceful degradation.
Economic attack	Supply ceilings, rate limits, concentration limits, delayed settlement and canary exposure.
Insider abuse	Separation of duties, immutable logs, least privilege, key rotation and independent incident review.
Privacy leakage	Data minimization, aggregate metrics, consent, retention limits and privacy-preserving eligibility proofs.

14.1 Emergency authority

A guardian multisignature may isolate a product or capability when an exploit, economic incident or harmful content event is credible. Guardians cannot transfer user assets, create supply or permanently change governance. Every action requires a public incident record, scope, evidence, expiry and DAO review. Phase 1 may use a Ztakora- controlled but multi-party 3-of-5 signer set; later phases should diversify signers and test replacement procedures.

14.2 Data policy

- Collect only telemetry required for gameplay, fraud control, reliability and agreed adoption measurement.
- Use aggregate or pseudonymous evidence wherever individual event histories are unnecessary.
- Publish retention periods and deletion processes for off-chain data.
- Separate identity eligibility credentials from public wallet activity where feasible.
- Do not expose private relationship or anomaly scores as public accusations without due process.

15 Operations, Maintenance and Recovery

Creation is a short event; operation is a continuing obligation. A company cannot graduate on the strength of a launch team alone. It must demonstrate support ownership, monitoring, release management, incident response and funded continuity.

REFERENCE TABLE

Requirement	Production expectation
Service owner	Named accountable operator and backup contact.
Service level	Availability, support response and recovery objectives proportionate to risk.
Monitoring	Contract events, adapter health, errors, latency, economic anomalies and asset availability.
Maintenance reserve	Restricted balance covering the published minimum operating period.
Release process	Versioning, compatibility tests, canary group, rollback and player notice.
Incident process	Detection, isolation, communication, remediation, compensation review and postmortem.
Continuation license	Pre-agreed right allowing another approved operator to maintain essential functionality after abandonment.
Retirement plan	Stop sales, preserve ownership evidence, migrate dependencies and communicate support end.

15.1 Recovery sequence

1. Issue an on-chain and in-client maintenance warning with the affected scope. 2. Restrict new minting, sales or high-risk actions while preserving ordinary customer access where safe. 3. Activate fallback providers or degraded mode for dependent products. 4. Offer the incumbent operator a bounded remediation period. 5. Open a continuation proposal to qualified replacement teams if the obligation is not restored. 6. Transfer only approved operating permissions and designated maintenance funds after review. 7. Retire the service cleanly if no safe continuation is available.

V

Delivery & Decision

Sections 16-20 / Roadmap, evidence gates, risks, worked example and executive decision



16 Implementation Roadmap

The protocol should be delivered as successive evidence packages rather than as a single public launch. Each phase has an explicit exclusion list and decision gate.

IMPLEMENTATION PHASES / OUTPUTS AND BOUNDARIES

Phase	Duration	Primary output	Boundary
A - Foundation & simulation	6-8 weeks	Protocol constitution, architecture, economy simulator, asset manifest, threat model, clickable product flow and parameter registry.	No public chain deployment, token or open funding.
B - Vertical slice	10-14 weeks	One playable region, Innovation District, Proposal Registry, Milestone Vault, Company Factory, Asset Registry and two controlled companies.	Private testers; closed game credits; no investor return.
C - Controlled creator pilot	8-12 weeks	External builder onboarding, verifier workflow, service/resource token factory, canary admission and recovery exercise.	Allowlisted teams and capped exposure.
D - Open proposal network	Conditional	Permissionless proposals, broader sandbox creation, expanded domains and transparent governance participation.	Canonical admission remains gated.
E - Mature economic network	Conditional	Multiple suppliers, diversified governance, validated legal investment lane and selected regional expansion.	Only after operational and legal evidence.

16.1 Vertical-slice scope

- One canonical region and one isolated Innovation District
- One settlement credit with no public secondary market
- One Protocol DAO simulation and bounded guardian multisig
- Open proposal publishing with formal qualification and anti-spam bond
- Company Contract, milestone escrow and revenue/maintenance routing
- Functional ERC-721/1155 asset registry with Game Asset Manifest
- Two experimental companies with different dependency and economy profiles
- One successful graduation and one deliberately exercised recovery path
- Security, economic, content and operational dashboards
- No cross-chain bridge, public protocol token or permissionless financial return

16.2 Required workstreams

REFERENCE TABLE

Workstream	Accountability
Game product	Core loop, region design, Innovation District, player UX and service integration.
Protocol engineering	Registries, factories, vaults, capability controls, adapters and governance.
Economy	Settlement model, sources/sinks, company templates, simulations and exposure limits.
3D/content	Style profiles, manifest schema, pipeline, performance budgets and moderation.
Security	Threat model, test strategy, key management, monitoring, incidents and recovery exercises.
Legal/compliance	Terms, token classification, backing/investment boundary, privacy and regional restrictions.
Operations	Builder onboarding, reviewer operations, support, company health and governance administration.

17 Measurement and Decision Gates

The program should measure whether the protocol produces enjoyable, interoperable and maintainable game value - not whether it creates the largest number of proposals, tokens or transactions.

REFERENCE TABLE

Outcome	Representative measures	Decision meaning
Player value	Prototype task completion, repeat voluntary use and satisfaction.	Products solve a real game need without excessive explanation.
Creation efficiency	Time and cost from qualified proposal to playable prototype.	Open production is faster or more diverse than studio-only production.
Quality	Admission pass rate after prototype iteration, defect rate and moderation findings.	Standards improve output without blocking all experimentation.
Economic health	Inflation, concentration, price stability, sink coverage and dependency diversity.	New companies do not destabilize progression or scarcity.
Security	Critical findings, exploit loss, pause frequency and recovery time.	Risk remains within approved exposure and incidents are containable.
Operations	Maintenance reserve coverage, support response and abandoned-product rate.	Products remain supported after initial launch.
Composability	Number of safe cross-product interactions and fallback coverage.	Companies expand the world rather than creating isolated NFT collections.
Governance quality	Participation diversity, decision time, conflict disclosure and appeal rate.	Governance remains legitimate without dominating gameplay.

17.1 Phase decision gate

- At least one third-party team can build a prototype using the published SDK and standards.
- At least one product demonstrates repeat voluntary player adoption without test rewards.
- A failed product can be isolated and recovered without loss of unrelated player assets.
- The economy remains inside declared inflation, concentration and progression limits during canary operation.
- The maintenance reserve and continuation process work in a live exercise.
- Security review finds no unresolved critical issue in foundation contracts or capability boundaries.
- Legal review confirms the proposed launch geography, funding flow and token treatment.
- Ordinary players can use the product without understanding DAO or smart-contract mechanics.

18 Consolidated Risk Register

Risk cannot be eliminated from an open game economy. The objective is to remove unnecessary risk, cap unavoidable exposure, detect failures early and preserve a credible recovery path.

CONSOLIDATED RISK REGISTER / ARCHITECTURAL RESPONSE

ID	Risk	Severity	Primary cause	Architectural response
R1	Game is not enjoyable	Critical	Economy/governance built before core play	Require a strong studio-owned core loop and player value gates before expansion.
R2	Fake adoption	Critical	Related accounts, paid cohorts or scripted activity	Unrewarded cohorts, identity attestations, anomaly review and independent challenge.
R3	Token speculation	Critical	Company tokens issued before utility	One settlement asset; utility-first factory; no default company ERC-20.
R4	Regulatory exposure	Critical	Backers expect financial return	Separate permissionless backing from regulated investment and legal entities.
R5	Smart contract exploit	Critical	Custom code and broad permissions	Audited templates, capability limits, canary exposure, timelock and targeted pause.
R6	Economic destabilization	Critical	Excess supply, progression bypass or monopoly	Simulation, caps, fallback suppliers, canary region and rollback thresholds.
R7	Governance capture	High	Purchased or permanent influence	Non-transferable decaying reputation, expiring delegation and conflict controls.
R8	DAO fatigue	High	Too many votes and organizations	One Protocol DAO; Company Contracts for routine operation.
R9	Inconsistent world quality	High	Unrestricted canonical assets	Innovation District, regional style profiles and creative admission review.
R10	Abandoned product	High	Team leaves after launch	Maintenance reserve, continuation license, replacement workflow and retirement.
R11	Dependency cascade	Critical	Critical service failure propagates	Adapters, circuit breakers, fallback mode, versioning and dependency health.
R12	IP infringement	High	Stolen or incompatible content	Provenance, license attestations, challenge/takedown and quarantine.
R13	Privacy harm	High	Identity and behavior over-collection	Data minimization, aggregate evidence, retention limits and due process.
R14	Early-user oligarchy	High	Permanent reputation and scarce control	Reputation decay, catch-up mechanics, supplier diversity and term limits.
R15	Scope failure	Critical	Attempt to build complete society at once	Vertical slice, explicit exclusions and evidence-based phase gates.

18.1 Residual-risk ownership

Every critical and high risk must have a named Ztakora owner, leading indicator, review cadence and stop condition before a public pilot. Smart contracts can enforce limits, but product, legal, security and creative accountability remains human and organizational.

19 Worked Example: Cargo Transportation Company

19.1 Need

Players in a growing city cannot move crafted goods between the industrial district and distant markets efficiently. Existing manual transport is repetitive and prevents regional commerce from scaling.

19.2 Proposal and prototype

A proposer publishes a cargo service concept. A team forms with a smart contract developer, game developer, economy designer, 3D vehicle artist and operator. Backers fund a limited prototype through a milestone vault. The team creates cargo containers, vehicles, stations, a route scheduler and fare settlement module inside the Innovation District.

19.3 PoUA trial

REFERENCE TABLE

Proof	Evidence
Need	Players document repeated transport friction and existing market abandonment.
Function	Users can book, load, track and collect cargo across two sandbox stations.
Adoption	Unrewarded merchants return to the service over multiple trade cycles.
Integrity	Qualifying activity excludes the builder team, backers and test incentives.
Safety	Fare, custody and cargo-release contracts pass defined tests and exposure caps.
Economic fit	The service reduces friction without making geography or local production irrelevant.
Maintainability	The operator funds a reserve and signs a continuation/retirement plan.

19.4 Canary and growth

The service enters one canonical route with a cap on daily cargo value, approved item categories and a fallback manual-claim process. If reliable and economically stable, it graduates. Its operation then creates credible new needs for vehicle maintenance, navigation, storage, cargo insurance alternatives, station retail and manufacturing. Those needs enter the same proposal-to-product process; they are not automatically deployed.

20 Recommended Ztakora Decision

RECOMMENDATION

Approve ZSEP v1.0 as the design baseline for architecture estimation and a controlled vertical slice. Do not authorize a public token, unrestricted Company DAOs, real-return crowdfunding, direct canonical deployment or cross-chain operation at this stage.

20.1 Immediate decisions

8. Confirm that the protocol's primary goal is useful game-world production, not token distribution or investment volume. 9. Select the initial game region and two company use cases with different economic and dependency profiles. 10. Appoint owners for game product, protocol engineering, economy, 3D/content, security, legal and operations. 11. Estimate Phase A and Phase B staffing, cost and chain requirements. 12. Draft the Protocol Constitution, PoUA metric classes and Company Contract permission model. 13. Run legal classification before selecting any externally transferable settlement or service token. 14. Define the failure exercise that the vertical slice must demonstrate before expansion.

20.2 Approval boundaries

Approval of this specification should authorize design, estimation, prototyping and controlled testing only. Public economic launch requires a separate decision package demonstrating player demand, security readiness, operational recovery, economic stability and jurisdiction-specific legal clearance.

VI

Reference Frameworks

Appendices A-D / Parameters, milestones, glossary and constitutional prohibitions



APPENDIX A

Initial Parameter Framework

REFERENCE TABLE

Parameter	Initial approach	Design intent
Formal proposal bond	Small and refundable	Spam deterrence, not access pricing.
Prototype grant ceiling	Risk-class dependent	Limit treasury exposure before a working build.
Innovation trial period	Product-cycle dependent	Long enough to observe voluntary repeat use.
Canonical canary	One region / bounded cohort	Limit value, geography, calls and dependencies.
Maintenance reserve	Several months of expected cost	Exact minimum determined by service criticality.
Protocol fee	Constitutional maximum	Should not make low-value services uneconomic.
Token supply	Fixed cap or bounded function	No unlimited administrator mint.
Governance delegation	Expiring	Requires active renewal and conflict disclosure.
Emergency pause	Scope-limited and expiring	Cannot transfer assets or create supply.
Upgrade timelock	Risk-class dependent	Longer for ownership and settlement components.

Numeric defaults should be set only after the initial game loop, expected player volumes, selected chain and operating cost model are known. Parameters must be recorded in a versioned registry and changed through the appropriate governance path.

APPENDIX B

Minimum Acceptance Package

REFERENCE TABLE

Package area	Required evidence
Product	Playable build, user journey, support material and known limitations.
Code	Source, reproducible build, dependency lock, tests, findings and deployment record.
Contracts	Addresses, roles, permissions, supply rules, upgrade path and emergency controls.
Economy	Sources/sinks, simulations, pricing, concentration, abuse cases and exposure limits.
Assets	Game Asset Manifests, licenses, provenance, performance and moderation results.

Package area	Required evidence
Adoption	Cohort definition, raw evidence references, exclusions, metrics and reviewer findings.
Operations	Owner, monitoring, maintenance reserve, incident and recovery procedures.
Legal	Terms, rights, restrictions, privacy treatment and any jurisdiction-specific clearance.
Dependencies	Providers, versions, fallbacks, health checks and customer impact.
Communication	Player disclosures, fees, rights, risks, support and planned changes.

APPENDIX C

Glossary

REFERENCE TABLE

Term	Definition
Canonical World	The stable player-facing world containing approved regions, systems and products.
Company Contract	A bounded on-chain charter defining one company's roles, permissions, treasury and lifecycle.
DAO	Decentralized Autonomous Organization; in ZSEP, the constitutional governance body for shared protocol rules.
Domain	A discovery category such as transport, commerce or housing; not automatically a separate DAO.
Functional NFT	A token referencing a usable, versioned game asset and its rights, not merely a collectible image.
Game Asset Manifest	The structured record of content, runtime, rights, provenance, safety and lifecycle data for an asset.
Innovation District	An isolated game region in which experimental products can operate without canonical economic access.
PoUA	Proof of Useful Adoption; the evidence-based admission process for expanded product permissions.
Protocol Constitution	The highest-level rules limiting governance, economics, security and product admission.
Service token	A capped representation of prepaid or consumable service utility, without default governance or equity rights.
Settlement asset	The single approved base asset or credit used for ordinary payments across the ecosystem.

APPENDIX D

Constitutional Prohibitions

- No uncapped emissions or unlimited owner minting.
- No reward for raw votes, wallet count, transaction count, asset count or validator volume.
- No self-validation or concealed acceptance tests.
- No permissionless direct deployment into the Canonical World.
- No guaranteed return, appreciation, redemption, yield or liquidity claim.
- No unrestricted sale of combat power or permanent progression bypass.
- No silent content replacement or retroactive rights reduction.
- No permanent protocol subsidy for an economically failed company.
- No unrestricted external calls or unbounded dependency exposure.
- No protocol-owned cross-chain bridge in the initial release.
- No single actor controlling proposal, treasury, verification and emergency intervention for the same product.

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